

Quasi-Elastic Neutron Scattering on Bimodal Dynamics in Molecular Solutions of Cellulose

Tao Xiong, Madhusudan Tyagi, Wei Zhou, Hongyu Guo, Robert M. Briber, and Howard Wang*



Cite This: *J. Phys. Chem. B* 2025, 129, 6714–6720



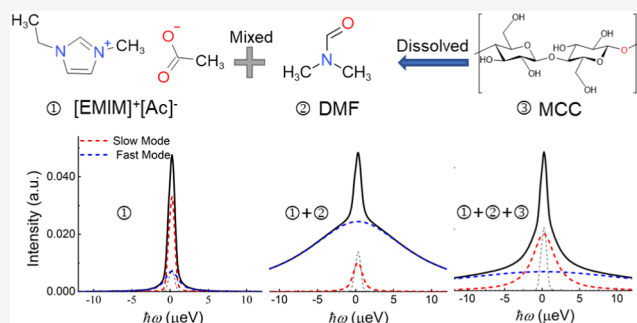
Read Online

ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: Local dynamics in molecular solutions of cellulose have been studied using quasi-elastic neutron scattering on ternary mixtures of microcrystalline cellulose, ionic liquid 1-ethyl-3-methylimidazolium acetate, and cosolvent dimethylformamide. Two distinct dynamic behaviors in solutions have been identified, with slow dynamics likely attributed to long-range translational diffusion and fast dynamics likely involving the motion of IL in nano-aggregates. The presence of cellulose suppresses both dynamic modes as the residence time for both slow and fast dynamics increases, while the radius of confinement remains invariant upon cellulose dissolution. Arrhenius activation behaviors are followed generally; the activation energy decreases for fast dynamics upon cellulose dissolution while showing no measurable change for slow ones. This study offers new information about the dynamics of an important class of renewable biomaterials.



INTRODUCTION

Cellulose is the most abundant renewable polymer on the Earth and holds great promise as an unexhaustive raw material for broad-range applications, from consumer products to building constructions to heavy industries in a sustainable circular economy. However, significant challenges remain in its widespread use, particularly due to the lack of safe, efficient, and cost-effective methods for processing cellulose from its natural form to final products. This is especially true for the step of dissolving cellulose from crystalline fibrils into molecular solutions, which involves the disruption of hydrogen bonding networks and London dispersion interactions from sugar packing in an often-heterogeneous local environment, giving rise to frustrated structures and diverse dynamics.^{1,2} Ionic liquids (ILs) are eco-friendly and considered promising solvents for cellulose dissolution for employing versatile mechanisms of cations and anions collaboratively complexing with glucose units.^{3,4} In a collective effort to improve the processes and reduce the cost to meet industrial demands, aprotic cosolvents, such as dimethyl sulfoxide (DMSO) and dimethylformamide (DMF), have been incorporated to enhance the efficiency of cellulose dissolution in ILs at lower temperatures.^{5–8}

Several studies have investigated the solution states of cellulose, the mechanism of dissolution, and optimal formulations.^{9–15} Cosolvents significantly reduce viscosity, improve solubility, and accelerate dissolution rates¹⁵ due to their preferential association with cations, which disrupt the strong ion–ion interactions found in ILs.¹¹ On the other hand,

both molecular dynamics simulations and quantum chemistry calculations suggest that the dissolution of cellulose in IL/cosolvent systems relies on hydrogen bond interactions between anions and the hydroxyl groups of cellulose. One study indicates that the presence of aprotic solvents stabilizes the solution by disrupting the aggregation of cellulose and anions,¹¹ while another study suggests that cosolvents primarily enhance the mass transport of ILs without altering cellulose structures.¹²

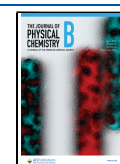
Small-angle neutron scattering (SANS) provides insights into the molecular structures of cellulose in ternary solutions.^{16–18} However, the dynamics of the dissolution system, although important, remain poorly understood. In a prior study,¹⁸ the structural partitioning of IL in cellulose solution was identified and semiquantitatively assessed. The study implies varying solvation layers of IL/solvent around cellulose chains as well as nanoscale micellar assemblies in solvents. The structural partition is relevant for dynamic segregation, as revealed by the temperature-dependent scan of the mean square displacement (MSD) in solutions. Two distinct dynamic transitions in cellulose solutions have been

Received: March 31, 2025

Revised: June 13, 2025

Accepted: June 16, 2025

Published: June 24, 2025



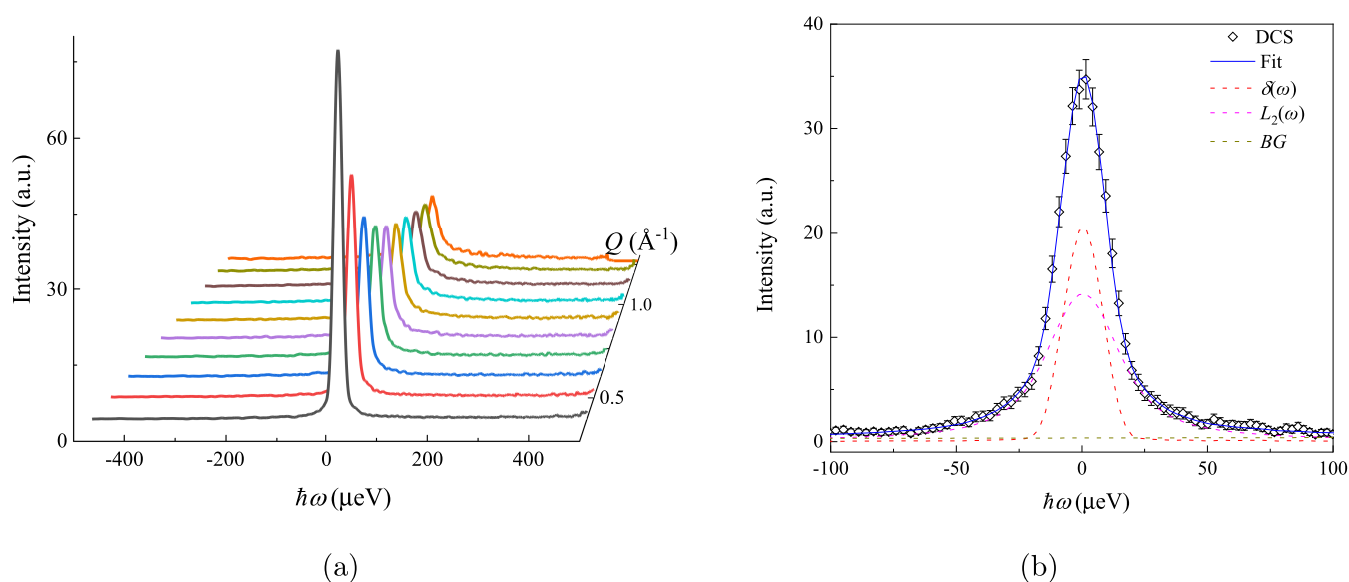


Figure 1. DCS spectra of the cellulose solution at 330 K. (a) Intensity spectra at various Q values from 0.4 to 1.2 $\text{\AA}^{-1}$. (b) Representative best model fitting of the spectrum at $Q = 0.50 \text{ \AA}^{-1}$ with component curves. Legend: open diamonds for experimental data, solid line curve for total intensity, dashed line curves for component intensity.

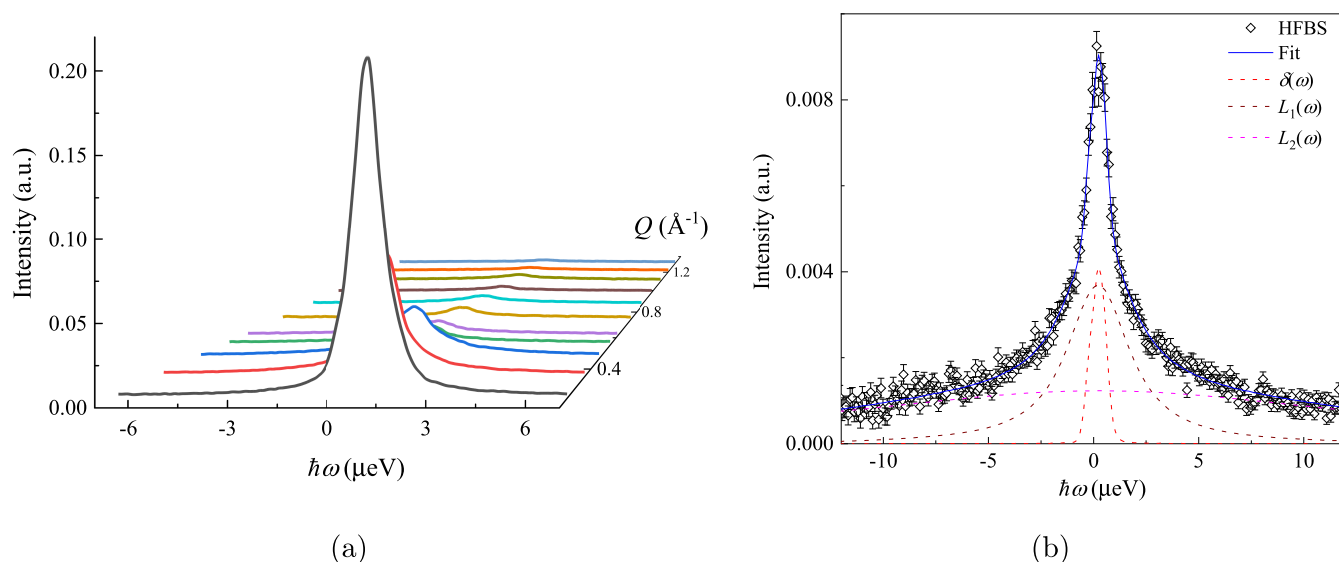


Figure 2. HFBS spectra of the cellulose solution at 300 K. (a) Intensity spectra at various Q values from 0.25 to 1.3 $\text{\AA}^{-1}$. (b) Representative best model fitting of spectrum at $Q = 0.75 \text{ \AA}^{-1}$ with component curves. Legend: open diamonds for experimental data, solid line curve for total intensity, dashed line curves for component intensity.

identified and attributed to the freezing/thawing of the molecular cellulose complexes and micellar assemblies, respectively.¹⁸ This study expands on the previous study through systematic measurements of dynamics in ternary cellulose solutions using quasi-elastic neutron scattering (QENS). The baseline QENS studies on the dynamics of neat IL and IL/solvent mixtures were also carried out.

METHODS

Microcrystalline cellulose (MCC) in catalog label Avicel PH101 and the ionic liquid 1-ethyl-3-methylimidazolium acetate (EMIMAc) were purchased from Sigma-Aldrich, and deuterated DMF (dDMF) was obtained from Cambridge Isotope. Solutions were prepared following the protocols as described in a previous study.¹⁸ Species is labeled based on the

molar percentage of cellulose (C), ionic liquid (I), and cosolvent (S), with C by moles of anhydroglucose units (AGU), and I and S by moles of molecules, respectively. The cellulose solution used in this study was prepared with a composition of 5% AGU, 15% IL, and 80% dDMF in molar percentage, labeled as C5I15S80. The molar ratio between cellulose and IL adheres to the asymptote $I/C = 3$.¹⁸ For comparative analysis, two additional samples were prepared: a solution consisting of 20% EMIMAc and 80% dDMF labeled as I20S80, and neat EMIMAc, labeled as I100.

QENS measurements were carried out using two spectrometers, the disk chopper spectrometer (DCS) and the high-flux backscattering spectrometer (HFBS) at the NIST Center for Neutron Research (NCNR) in Gaithersburg, Maryland. DCS is a direct geometry time-of-flight spectrometer and used

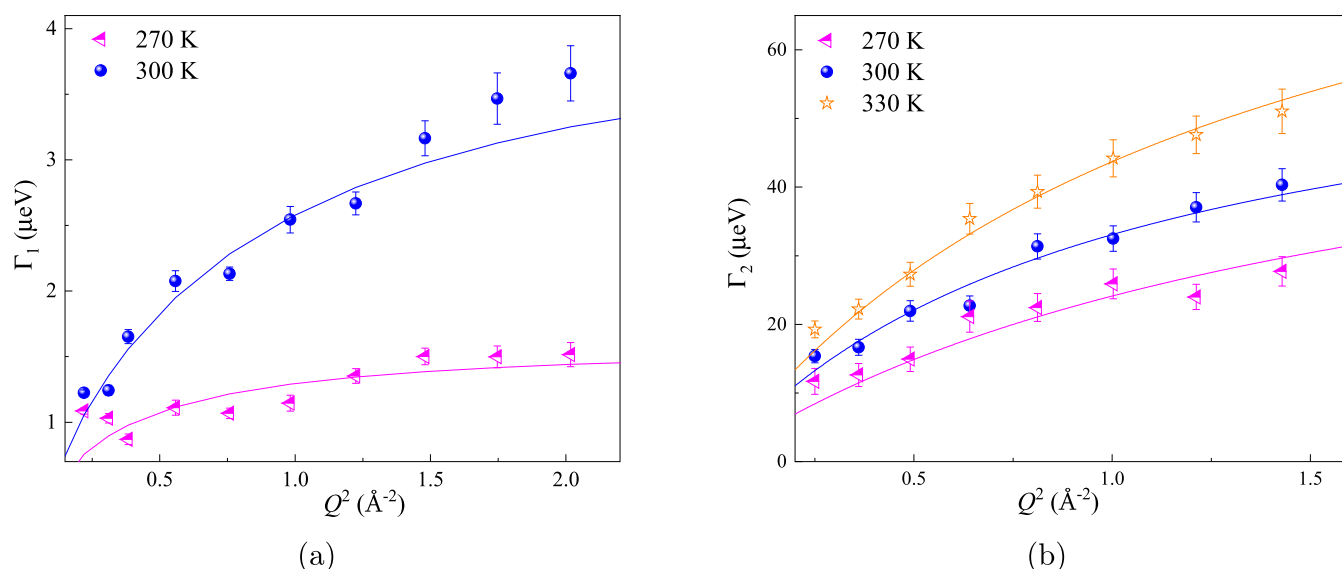


Figure 3. Q -dependent Γ_i of cellulose solution at different temperatures for (a) slow dynamics and (b) fast dynamics (purple triangles: 270 K; blue circles: 300 K; orange pentagrams: 330 K; and solid line curves through symbols are best fitting using the jump diffusion model).

in a configuration with an energy resolution of ca. 19 μeV , covering dynamics range from -0.5 to 0.5 meV, with the amplitude of the momentum transfer vector Q , ranging from 0.22 to 1.25 $\text{\AA}^{-1}$.¹⁹ HFBS offers a high energy resolution of 0.8 μeV with a dynamics range of -17 to 17 μeV and a Q range spanning from 0.25 to 1.75 $\text{\AA}^{-1}$,²⁰ covering slower dynamics not readily accessible with DCS. Samples were sealed in hollow containers for the experiments, and a sample thickness of 0.5 mm was chosen to ensure about 10% of scattering to avoid multiple scattering issues. All solutions were measured on DCS at 270, 300, and 330 K. For experiments on HFBS, CS115S80 and I100 were measured at 270 and 300 K, while I20S80 was measured at 240 and 270 K. The resolution function was determined by measuring vanadium and the neat IL at 4 K. Data processing was conducted using DAVE software package.²¹

RESULTS AND DISCUSSION

The DMF is deuterated, and the number of hydrogen atoms in the IL is four times greater than that in cellulose. Thus, the

Table 1. Diffusion Parameters of Cellulose Solution at Different Temperatures

temp	D_1 ($10^{-11}\text{m}^2 \cdot \text{s}^{-1}$)	τ_{01} (ps)	D_2 ($10^{-11}\text{m}^2 \cdot \text{s}^{-1}$)	τ_{02} (ps)
330 K			116 ± 7	7 ± 1
300 K	10 ± 2	152 ± 17	93 ± 5	9 ± 1
270 K	6 ± 1	338 ± 24	58 ± 8	10 ± 2

signals of QENS experiments primarily reflect the mobility of the IL within the sample. The spectra of the cellulose solution obtained from DCS and HFBS are illustrated in Figures 1a and 2a, respectively. The QENS spectra broaden as Q increases, implying a process of translational diffusion.

Typical QENS spectra compound elastic contributions from the slower processes that the instruments cannot resolve. An empirical approach to account for the elastic contribution is to allow a variable elastic term in the fit model.^{22–24} DCS spectra are modeled satisfactorily as the sum of a Lorentzian and an

elastic Gaussian function with a linear background, expressed as follows:

$$S(Q, \omega) = b_1(Q)\delta(\omega) + b_2(Q)L_2(\omega) + \text{BG} \quad (1)$$

The Lorentzian function $L_2(\omega)$ represents the fast dynamics of IL. The best fitting of the DCS spectrum at $Q = 0.50$ $\text{\AA}^{-1}$ is shown in Figure 1b.

For HFBS spectra, a model consisting of two Lorentzian functions and an elastic component with a negligible background has yielded satisfactory fitting, implying a convolution of fast and slow dynamics, expressed as follows:

$$S(Q, \omega) = a_1(Q)\delta(\omega) + (1 - a_1(Q))S_1(Q, \omega) \times S_2(Q, \omega) \quad (2)$$

The above equation and subsequent equations in this manuscript illustrate model scattering functions. For actual data fitting, these functions are convoluted with respective instrument resolutions, as the collected data are broadened due to instrument resolution.

In previous QENS studies of neat IL and IL/water mixtures, two distinct dynamics have been identified.^{22,23,25,26} The slow dynamics is attributed to long-range translational diffusion, while the fast dynamics results from the motion of IL within the aggregates. Evidence of mesoscopic aggregation in imidazolium-based ILs has been obtained using dielectric relaxation and optical Kerr effect,²⁷ and supported by X-ray diffraction.²⁵ The micellar complexation of IL with cosolvent has been illustrated using SANS and NMR.^{15,28}

Similarly, the dynamics structure factor for the long-range diffusion, $S_1(Q, \omega)$, and the confined diffusion, $S_2(Q, \omega)$, are expressed as

$$S_1(Q, \omega) = \frac{1}{\pi} \frac{\Gamma_1(Q)}{\omega^2 + \Gamma_1(Q)^2} \quad (3)$$

and

$$S_2(Q, \omega) = p(Q)\delta(\omega) + (1 - p(Q))\frac{1}{\pi} \frac{\Gamma_2(Q)}{\omega^2 + \Gamma_2(Q)^2} \quad (4)$$

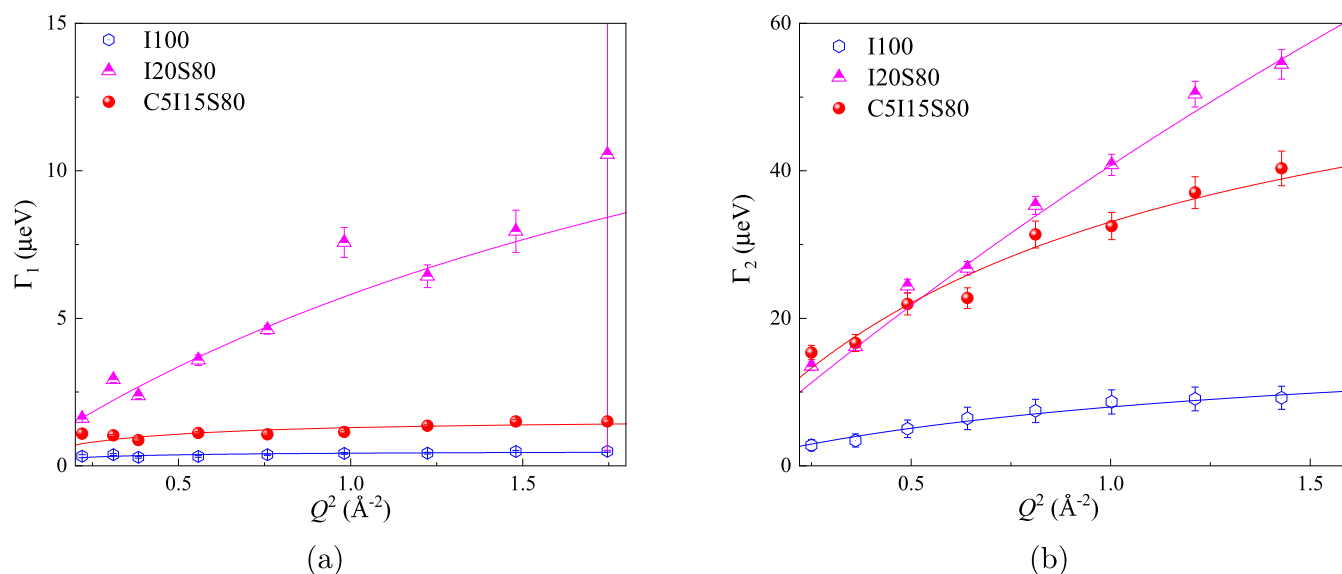


Figure 4. Comparison of Γ_i for three samples for (a) slow dynamics, 270 K, and (b) fast dynamics, 300 K. Blue hexagons: I100; purple triangles: I20S80; red circles: C5I15S80; and solid line curves are best fitting using the jump diffusion model.

Table 2. Diffusion Parameters of Three Samples for Slow and Fast Dynamics

sample	slow dynamics, 270 K			fast dynamics, 300 K		
	D_1 ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)	τ_{01} (ps)	$\langle L_i^2 \rangle / 2$ (Å)	D_2 ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)	τ_{02} (ps)	$\langle L_i^2 \rangle / 2$ (Å)
I100	4 ± 1	1338 ± 114	5.7 ± 0.8	39 ± 3	16 ± 2	1.9 ± 0.2
I20S80	12 ± 2	31 ± 12	1.5 ± 0.4	70 ± 4	2 ± 1	0.9 ± 0.2
C5I15S80	6 ± 1	473 ± 43	3.5 ± 0.4	93 ± 5	9 ± 1	2.2 ± 0.2

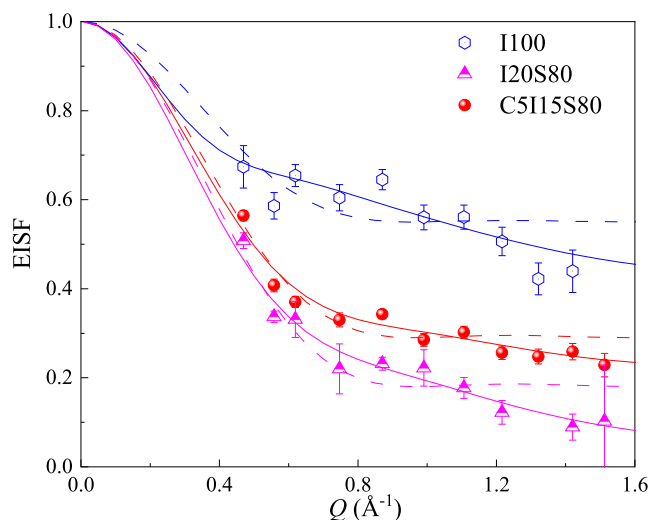


Figure 5. Q -dependence of the EISF at 270 K for three samples. Blue hexagons: I100; purple triangles: I20S80; red circles: C5I15S80; and the solid and dashed line curves through symbols are the best fitting with and without the correction terms as in eq 8, respectively.

Table 3. Parameters Obtained by the Fits of EISF

sample	c_1	a_1 (Å)	c_2	a_2 (Å)
I100	0.7 ± 0.1	8 ± 2	0.5 ± 0.1	1.3 ± 0.3
I20S80	0.3 ± 0.1	5.0 ± 0.6	0.02 ± 0.2	1.3 ± 0.2
C5I15S80	0.4 ± 0.1	5.1 ± 0.4	0.5 ± 0.1	1.3 ± 0.4

respectively, where $\Gamma_1(Q)$ and $\Gamma_2(Q)$ are the half-widths at half-maximum (HWHM) of the Lorentzian for long-range and

confined diffusion, respectively. $p(Q)$ denotes the fraction of “immobile” species in the confined diffusion, also known as the elastic incoherent structure factor (EISF). By incorporating eqs 3 and 4, the fitting function can be expressed as

$$S(Q, \omega) = a_1(Q)\delta(\omega) + a_2(Q)L_1(\omega) + a_3(Q)L_2(\omega) \quad (5)$$

where the delta function accounts for the elastic background. Then, the EISF can be expressed as

$$p(Q) = \frac{a_2(Q)}{a_2(Q) + a_3(Q)} \quad (6)$$

The broader Lorentzian $L_2(\omega)$ is insensitive to the HWHM due to the narrow dynamics range of HFBS, so the HWHM is fixed based on the results of DSC during the fitting procedure. The fit result at $Q = 0.75 \text{ Å}^{-1}$ obtained from HFBS is shown in Figure 2b. The control samples, consisting of neat EMIMAc and a mixture of EMIMAc and dDMF, were modeled by following the same method.

The peak widths $\Gamma_i(Q)$ of the cellulose solution at different temperatures are shown in Figure 3. The $\Gamma_i(Q)$ values increase with both the temperature and Q , eventually reaching a plateau at high Q , indicating a non-Fickian diffusion process. The $\Gamma_i(Q)$ data are typically assessed using the jump diffusion model:

$$\Gamma_i(Q) = \frac{\hbar D_i Q^2}{1 + D_i Q^2 \tau_{0i}} \quad \text{for } i = 1, 2 \quad (7)$$

where D_i denotes the diffusion coefficient and τ_{0i} represents the average residence time between jumps. The diffusion coefficients and residence times obtained from the jump

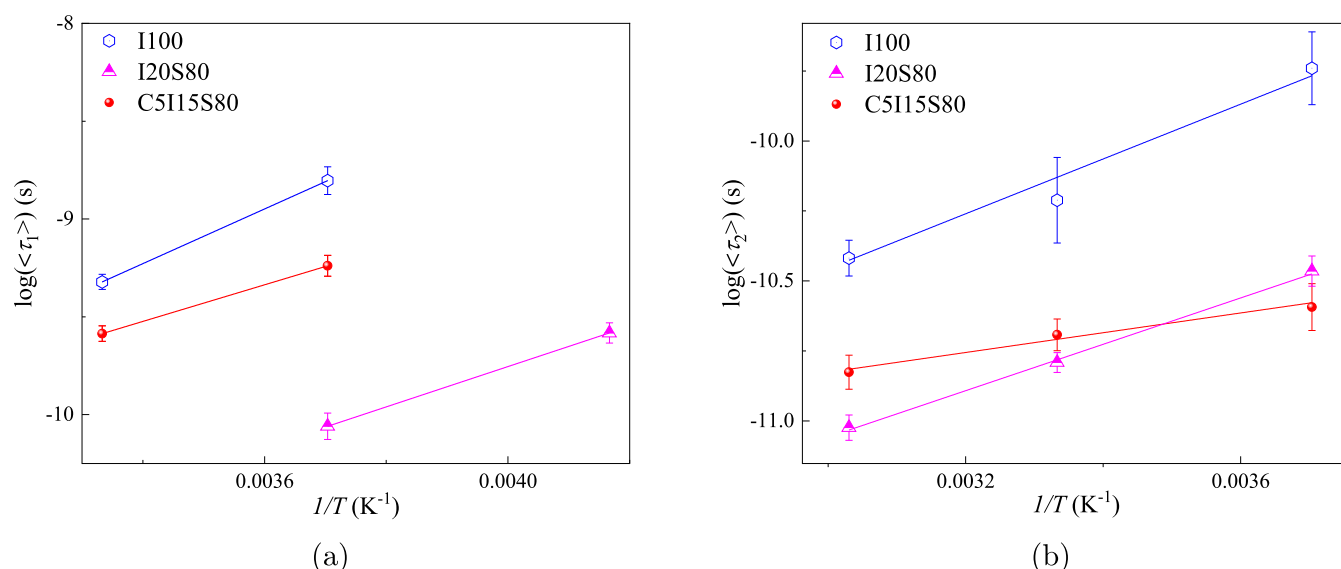


Figure 6. Arrhenius plots of relaxation times for (a) slow dynamics and (b) fast dynamics. Blue hexagons: I100; purple triangles: I20S80; red circles: C5I15S80; and solid line curves through symbols are best fitting using the Arrhenius equation.

Table 4. Activation Energies for the Slow (E_{a1}) and Fast (E_{a2}) Dynamics in Three Samples

sample	E_{a1} (kJ · mol ⁻¹)	E_{a2} (kJ · mol ⁻¹)
I100	12 ± 2	8.1 ± 1.8
I20S80	9 ± 2	6.9 ± 0.3
C5I15S80	9 ± 2	2.9 ± 0.5

diffusion model at different temperatures are given in Table 1. The diffusion coefficients increase with temperature for both dynamics, while the residence time decreases.

A comparison of $\Gamma_i(Q)$ of three samples is shown in Figure 4a for slow dynamics at 270 K and in Figure 4b for fast dynamics at 300 K, respectively. The symbols in the figures are obtained from fitting the experimental data, while the solid line curves represent the best fit using the jump diffusion model. $\Gamma_i(Q)$ of I20S80 is significantly larger than those of I100 for both slow and fast dynamics, indicating the dilution effect on dynamics with the addition of cosolvent. Upon the dissolution of cellulose, $\Gamma_1(Q)$ values of C5I15S80 are significantly smaller than those of I20S80, implying the suppression of slow dynamics, likely due to the immobilization of IL upon adsorption on cellulose. On the other hand, $\Gamma_2(Q)$ values of C5I15S80 are similar to those of I20S80, implying that the presence of cellulose has insignificant effects on C5I15S80, similar to those on I20S80, implying insignificant effects of the presence of cellulose on micellar assemblies of IL in solutions.

The corresponding diffusion coefficients and residence times obtained from the fitting, as shown in Figure 4 for three samples, are listed in Table 2. The diffusion coefficients of IL increase with the addition of cosolvent in the neat IL for both dynamics, while the residence times decrease. In the cellulose solution, the presence of cellulose suppresses the IL dynamics as shown in the significant increase in residence times for both slow and fast dynamics, while the mean diffusion jump lengths ($\langle L_i^2 \rangle^{1/2}$) also increase for both. The overall effects are a smaller diffusion coefficient for slow dynamics at 270 K and a slightly larger one for fast dynamics at 300 K. As the increase in the residence time upon cellulose dissolution is expected, the increase in the mean jump length is counterintuitive. One

hypothesis is the symmetry breaking and confinement effect on the transport among rod-like 1D cellulose molecules. More measurements and new models are needed to test the hypothesis.

The Q -dependence of EISF at 270 K, obtained from eq 8, is shown in Figure 5. The data shown by solid symbols are fitted by using a localized diffusion model. To improve the fit, a contribution from localized process is included,²² with the following expression:

$$p(Q) = \left[c_1 + (1 - c_1) \left(\frac{3j_1(Qa_1)}{Qa_1} \right)^2 \right] \left[c_2 + (1 - c_2) j_0^2(Qa_2) \right] \quad (8)$$

where j_0 and j_1 denote the spherical Bessel functions of the zeroth and the first order, respectively. The parameters c_1 and a_1 correspond to the fast dynamics, with c_1 representing the “immobile” fraction and a_1 indicating their effective confinement radius. The parameters c_2 and a_2 in the second bracket are the correction terms that account for the local rotational dynamics within a confinement radius. The fit result is shown in Figure 5. When the correction term is excluded, there is a noticeable deviation at a higher Q , while the model with the correction fits the data well. The correction term may be attributed to the spatially constrained motion of side groups. The relevant parameters are listed in Table 3. The value of a_2 is close to the C–H length of the side group. The addition of cosolvent in the IL decreases the confinement radius of fast dynamics (a_1), while the presence of cellulose has little effect on a_1 . The spatial confinement of the fast process is supposedly indicated by a low- Q plateau in Γ_2 of fast dynamics. However, no obvious plateau is observed for all three samples. This may be because the reliable DCS data below $Q^2 = 0.3 \text{ \AA}^{-2}$ were not obtained, and the deviations of the fits, especially below $3.3/a_1$, which will be at low- Q side, are not easily visible in the data shown. There was also no plateau observed at low Q for faster process in similar research of ionic liquid [Im][TFSI] by DCS.²⁹ Symmetry breaking in transport may also play a role here.

The temperature dependence of the relaxation time at 1.0 Å, which is related to the peak width via $\langle \tau_i \rangle^{-1} = \Gamma_i / \hbar$, is shown in Figure 6. The Arrhenius model has been shown to be a reliable framework to describe the translational diffusion in similar ionic liquid systems.^{22,30,31} The activation energy of imidazolium-based ILs for the diffusive process is generally found to be about 18–23 kJ·mol⁻¹.³⁰ To estimate the activation energies (E_a) of the three samples, the relaxation time is fitted with the Arrhenius equation $\tau_i = A \exp(E_{ai}/k_B T)$. The estimated activation energies are listed in Table 4. The activation energy for the slow dynamics is greater than that for the fast dynamics for all three samples. The addition of the cosolvent to the neat IL has only marginal effect on the activation energies for both types of dynamics. Upon cellulose dissolution, the activation energy for the fast dynamics, E_{a2} , decreases from 6.9 kJ·mol⁻¹ in the IL/cosolvent solution to 2.9 kJ·mol⁻¹ in the cellulose solution. In contrast, activation energies for the slow dynamics, E_{a1} , remain unchanged.

CONCLUSIONS

We investigated the local dynamics of a ternary solution consisting of cellulose, an ionic liquid, and an organic cosolvent using quasi-elastic neutron scattering. For comparison, we also measured the neat IL as well as the mixture of IL/cosolvent. We identified two distinct dynamic modes for IL in solutions, with slow dynamics likely attributed to long-range translational diffusion and fast dynamics likely involving the motion of IL in nanoaggregates in solutions. The presence of cellulose increases the residence time as well as the mean jump length, resulting in opposite trends of the diffusion coefficient change, becoming smaller for slow dynamics and larger for fast dynamics. The activation energy becomes lower upon cellulose dissolution for fast dynamics while showing no measurable change for slow dynamics. Data in this study provide relevant information about the dynamics of cellulose solutions and suggest the effect of symmetry breaking and confinement on molecular transport. Future studies are needed to test the hypothesis.

AUTHOR INFORMATION

Corresponding Author

Howard Wang – Department of Materials Science and Engineering, University of Maryland, College Park, Maryland 20742, United States; orcid.org/0000-0001-9905-4049; Email: wangh@umd.edu

Authors

Tao Xiong – Institute of High Energy Physics, Chinese Academy of Sciences, Beijing 100049, China; Spallation Neutron Source Science Center, Dongguan, Guangdong 523803, China; orcid.org/0000-0002-9484-4903

Madhusudan Tyagi – NIST Center for Neutron Research, National Institute of Standards and Technology, Gaithersburg, Maryland 20899, United States; Department of Materials Science and Engineering, University of Maryland, College Park, Maryland 20742, United States

Wei Zhou – NIST Center for Neutron Research, National Institute of Standards and Technology, Gaithersburg, Maryland 20899, United States

Hongyu Guo – Institute of High Energy Physics, Chinese Academy of Sciences, Beijing 100049, China; Spallation Neutron Source Science Center, Dongguan, Guangdong 523803, China

Robert M. Briber – Department of Materials Science and Engineering, University of Maryland, College Park, Maryland 20742, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jpcb.5c02179>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

H.W. and R.M.B. acknowledge the support of the National Institute of Standards and Technology through award 70NANB12H238. Access to the HFBS was provided by the Center for High Resolution Neutron Scattering, a partnership between the National Institute of Standards and Technology and the National Science Foundation under Agreement No. DMR-2010792. The identification of any commercial product or trade name does not imply endorsement or recommendation by NIST.

REFERENCES

- (1) Li, Y.; Wang, J.; Liu, X.; Zhang, S. Towards a molecular understanding of cellulose dissolution in ionic liquids: anion/cation effect, synergistic mechanism and physicochemical aspects. *Chem. Sci.* **2018**, *9*, 4027–4043.
- (2) Li, Y.; Yan, C.; Chen, Y.; Han, X.; Shao, Z.; Qi, H.; Li, X.; Nishiyama, Y.; Hu, T.; Chen, P. The major role of London dispersion interaction in the assembly of cellulose, chitin, and chitosan. *Cellulose* **2023**, *30*, 8127–8138.
- (3) Swatoski, R. P.; Spear, S. K.; Holbrey, J. D.; Rogers, R. D. Dissolution of Cellose with Ionic Liquids. *J. Am. Chem. Soc.* **2002**, *124*, 4974–4975.
- (4) Raghuvanshi, V. S.; Cohen, Y.; Garnier, G.; Garvey, C. J.; Russell, R. A.; Darwish, T.; Garnier, G. Cellulose Dissolution in Ionic Liquid: Ion Binding Revealed by Neutron Scattering. *Macromolecules* **2018**, *51*, 7649–7655.
- (5) Rinaldi, R. Instantaneous dissolution of cellulose in organic electrolyte solutions. *Chem. Commun.* **2011**, *47*, 511–513.
- (6) Wu, L.; Lee, S.-H.; Endo, T. Effect of dimethyl sulfoxide on ionic liquid 1-ethyl-3-methylimidazolium acetate pretreatment of eucalyptus wood for enzymatic hydrolysis. *Bioresour. Technol.* **2013**, *140*, 90–96.
- (7) Holding, A. J.; Parviainen, A.; Kilpeläinen, I.; Soto, A.; King, A. W. T.; Rodríguez, H. Efficiency of hydrophobic phosphonium ionic liquids and DMSO as recyclable cellulose dissolution and regeneration media. *RSC Adv.* **2017**, *7*, 17451–17461.
- (8) El Seoud, O. A.; Kostag, M.; Jedvert, K.; Malek, N. I. Cellulose in Ionic Liquids and Alkaline Solutions: Advances in the Mechanisms of Biopolymer Dissolution and Regeneration. *Polymers* **2019**, *11*, 1917.
- (9) Xu, A.; Zhang, Y.; Zhao, Y.; Wang, J. Cellulose dissolution at ambient temperature: role of preferential solvation of cations of ionic liquids by a cosolvent. *Carbohydr. Polym.* **2013**, *92*, 540–544.
- (10) Zhao, Y.; Liu, X.; Wang, J.; Zhang, S. Insight into the Cosolvent Effect of Cellulose Dissolution in Imidazolium-Based Ionic Liquid Systems. *J. Phys. Chem. B* **2013**, *117*, 9042–9049.
- (11) Andanson, J.-M.; Bordes, E.; Devémy, J.; Leroux, F.; Pádua, A. A. H.; Gomes, M. F. C. Understanding the role of co-solvents in the dissolution of cellulose in ionic liquids. *Green Chem.* **2014**, *16*, No. 2528.
- (12) Radhi, A.; Le, K. A.; Ries, M. E.; Budtova, T. Macroscopic and microscopic study of 1-ethyl-3-methyl-imidazolium acetate-DMSO mixtures. *J. Phys. Chem. B* **2015**, *119*, 1633–1640.
- (13) Phadagi, R.; Singh, S.; Hashemi, H.; Kaya, S.; Venkatesu, P.; Ramjugernath, D.; Ebenso, E. E.; Bahadur, I. Understanding the role of Dimethylformamide as co-solvents in the dissolution of cellulose in

ionic liquids: Experimental and theoretical approach *J. Mol. Liq.* **328** 115392.

(14) Zhou, Y.; Zhang, X.; Yin, D.; Zhang, J.; Mi, Q.; Lu, H.; Liang, D.; Zhang, J. The solution state and dissolution process of cellulose in ionic-liquid-based solvents with different hydrogen-bonding basicity and microstructures. *Green Chem.* **2022**, *24*, 3824–3833.

(15) Minnick, D. L.; Flores, R. A.; DeStefano, M. R.; Scurto, A. M. Cellulose Solubility in Ionic Liquid Mixtures: Temperature, Cosolvent, and Antisolvent Effects. *J. Phys. Chem. B* **2016**, *120*, 7906–7919.

(16) Rein, D. M.; Khalfin, R.; Szekely, N.; Cohen, Y. True molecular solutions of natural cellulose in the binary ionic liquid-containing solvent mixtures. *Carbohydr. Polym.* **2014**, *112*, 125–133.

(17) Napso, S.; Rein, D. M.; Khalfin, R.; Cohen, Y. Semidilute solution structure of cellulose in an ionic liquid and its mixture with a polar organic co-solvent studied by small-angle X-ray scattering. *J. Polym. Sci. B: Polym. Phys.* **2017**, *55*, 888–894.

(18) Zhang, X.; Mao, Y.; Tyagi, M.; Jiang, F.; Henderson, D.; Jiang, B.; Lin, Z.; Jones, R. L.; Hu, L.; Briber, R. M.; et al. Molecular partitioning in ternary solutions of cellulose. *Carbohydr. Polym.* **2019**, *220*, 157–162.

(19) Copley, J. R. D.; Cook, J. C. The Disk Chopper Spectrometer at NIST: a new instrument for quasielastic neutron scattering studies. *Chem. Phys.* **2003**, *292*, 477–485.

(20) Meyer, A.; Dimeo, R. M.; Gehring, P. M.; Neumann, D. A. The high-flux backscattering spectrometer at the NIST Center for Neutron Research. *Rev. Sci. Instrum.* **2003**, *74*, 2759–2777.

(21) Azuah, R. T.; Kneller, L. R.; Qiu, Y.; Tregenna-Piggott, P. L. W.; Brown, C. M.; Copley, J. R. D.; Dimeo, R. M. DAVE: A Comprehensive Software Suite for the Reduction, Visualization, and Analysis of Low Energy Neutron Spectroscopic Data. *J. Res. Natl. Inst. Stand. Technol.* **2009**, *114*, 341–358.

(22) Mamontov, E.; Luo, H.; Dai, S. Tetramethylguanidinium Bis(perfluoroethylsulfonyl)-imide Protic Ionic Liquid Probed by Quasielastic Neutron Scattering. *J. Phys. Chem. B* **2009**, *113*, 159–169.

(23) Mamontov, E.; Baker, G. A.; Luo, H.; Dai, S. Microscopic Diffusion Dynamics of Silver Complex-Based Room Temperature Ionic Liquids Probed by Quasielastic Neutron Scattering. *ChemPhysChem* **2011**, *12*, 944–950.

(24) Mamontov, E.; Smith, R. W.; Billings, J. J.; Ramirez-Cuesta, A. J. Simple analytical model for fitting QENS data from liquids. *Phys. B* **2019**, *566*, 50–54.

(25) Ferdeghini, F.; Berrod, Q.; Zanolli, J. M.; Judeinstein, P.; Sakai, V. G.; Czakkel, O.; Fouquet, P.; Constantin, D. Nanostructuring of ionic liquids: impact on the cation mobility. A multi-scale study. *Nanoscale* **2017**, *9*, 1901–1908.

(26) Ruiz-Martín, M. D.; Qureshi, N.; González, M. A.; J Ollivier, B. F.; Farago, B. Influence of water on the microscopic dynamics of 1-butyl-3-methylimidazolium tetrafluoroborate studied by means of quasielastic neutron scattering. *J. Chem. Phys.* **2022**, *156*, No. 084505.

(27) Turton, D. A.; Hunger, J.; Stoppa, A.; Hefter, G.; Thoman, A.; Walther, M.; Buchner, R.; Wynne, K. Dynamics of Imidazolium Ionic Liquids from a Combined Dielectric Relaxation and Optical Kerr Effect Study: Evidence for Mesoscopic Aggregation. *J. Am. Chem. Soc.* **2009**, *131*, 11140–11146.

(28) Gao, J.; Wagner, N. J. Water Nanocluster Formation in the Ionic Liquid 1-Butyl-3-methylimidazolium Tetrafluoroborate ([C4mim][BF4])-D₂O Mixtures. *Langmuir* **2016**, *32*, 5078–5084.

(29) Hoarfrost, M. L.; Tyagi, M.; Segalman, R. A.; Reimer, J. A. Proton hopping and long-range transport in the protic ionic liquid [Im][TFSI], probed by pulsed-field gradient NMR and quasi-elastic neutron scattering. *J. Phys. Chem. B* **2012**, *116*, 8201–8209.

(30) Kofu, M.; Tyagi, M.; Inamura, Y.; Miyazaki, K.; Yamamuro, O. Quasielastic neutron scattering studies on glass-forming ionic liquids with imidazolium cations. *J. Chem. Phys.* **2015**, *143*, No. 234502.

(31) González, M. A.; Aoun, B.; Price, D. L.; Izaola, Z.; Russina, M.; Ollivier, J.; Saboungi, M.-L. Molecular dynamics in 1-alkyl-3-

methylimidazolium bromide ionic liquids: A reanalysis of quasielastic neutron scattering results. *AIP Confer. Proc.* **2018**, *1969*, No. 020002.



CAS BIOFINDER DISCOVERY PLATFORM™

ELIMINATE DATA SILOS. FIND WHAT YOU NEED, WHEN YOU NEED IT.

A single platform for relevant, high-quality biological and toxicology research

Streamline your R&D

CAS
A division of the American Chemical Society